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Module 4: Software Design
Session 6: Design Review and Stakeholder Presentation

Software Developer Level 4

What you will be able to do

By the end of this session you will be able to:

1. Apply the module's design techniques to extend the design of your own project or the fallback brief.
2. Evaluate a peer's design against agreed review criteria and give specific feedback.
3. Justify your design decisions to technical and non-technical stakeholders.

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Before we start

FROM YOUR PREPARE

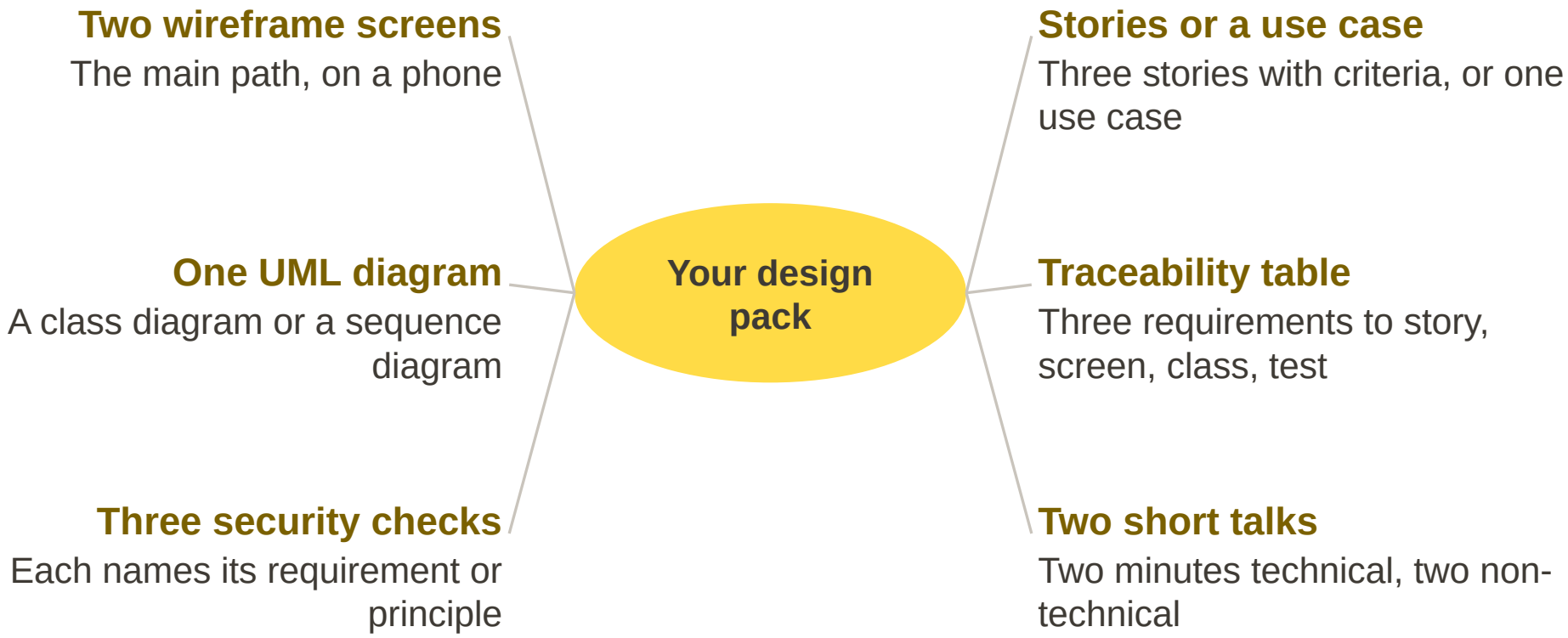
Which part of your design would you most like someone else to challenge?

Type one line in the chat: the part, and why it worries you.

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What goes in your design pack

One feature, explained well enough for someone else to review, build and test it.



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Choose your project now

Type your choice in the chat in one line, for example 'Shift swap in our rota app'.

Your own project (best)

Your Software Foundations project plan, or a feature you are building at work.

One feature you can design in 45 minutes: one journey, one screen, three requirements.

Describe it from what you know, never from real records.

Or the fallback brief

Case Brief Part 6: the repair satisfaction survey for Kestrel Moor Homes.

Use it if you have no project, or if yours is too sensitive to show.

Everyone designs it their own way: there is no single right answer.

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Keep it confidential

BEFORE YOU OPEN YOUR PROJECT

> **Design from your project, never from its data.**

No personal or customer data, credentials or confidential material in your pack or your talk.

No employer data in AI tools your employer has not approved.

Need example values? Invent obviously fake ones, such as 'Customer 0001'.

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How design reviews work in teams

Why teams review designs before building, who does what, and what reviewers look for.

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Why teams review a design before building it

A diagram is cheap to change
Moving a box takes a minute. After the build, the same change means rework, retesting and release.

Everyone builds the same thing
Developers, testers and the people who will use it agree what the design promises.

Gaps show up early
A missing requirement, a test nobody can write, data that should not be there.



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Who does what in a design review

Small teams combine roles. On your board, the timekeeper facilitates.

Role	What they do
Author	Presents the design, answers questions, decides what to change and owns the changes
Reviewers	Each looks through one lens: users, data, structure or the people who operate it
Facilitator	Keeps time and keeps the talk on the design, not the designer
Scribe	Records each point and who will act on it
Decision-maker	Agrees the design can move on, for example Kestrel Moor's Head of Repairs

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Seven review criteria

Reviewers judge the design against these, not against their own taste.

1. **Clarity**
   Could a new team member follow it without you?
2. **Consistency**
   One name for each thing, in every artefact?
3. **Completeness**
   Error and edge paths, not only the happy path?
4. **Traceability**
   Can each requirement be followed to a test?
5. **Accessibility**
   Labels, errors in text, contrast, targets, plain English?
6. **Security**
   Personal data minimised, input checked, failures loud?
7. **Communication**
   Does each audience get the story it needs?

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Safe to be wrong

HOW EFFECTIVE TEAMS WORK

> **A review only finds problems if people feel safe to raise them.**

Edmondson (1999) called this psychological safety: a shared belief that the team is safe for interpersonal risk taking.

So review the design, never the designer, and thank people for the problems they find.

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Feedback the designer can act on

Example: the contact preferences screen from Session 2.

VAGUE

'The preferences screen is a bit confusing.'

Where? No place named

Confusing how? No criterion

What now? The designer has to guess

SPECIFIC

'The text and email toggles have no visible labels. Add one to each.'

Where: the two toggles

Criterion: accessibility (visible labels)

Change: one the designer can make today

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Owning your design pack

Today you work on your own design. Nobody else can defend it for you.

You decide

Which feature to design, and what to leave out

Which diagram explains it best

When it is good enough to show

You answer for

Every link in your traceability table

The risks you have not solved yet, named out loud

What you change after the review, and why

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One design, two talks

Adapting the same design for technical and non-technical audiences.

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Same design, different questions

ADAPTING TO YOUR AUDIENCE

> **The design stays the same; the story changes with the audience.**

A developer asks: how does it work, and what could break?

A tenant asks: what changes for me, and what do you need from me?

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The notifications design, told twice

Worked example: the Session 5 notifications design. About two minutes each.

TECHNICAL TALK: FOR DEVELOPERS

'A new channel is one new class; repair logic never changes.'

Status and appointment events trigger NotificationService (Observer)

Text, email and letter share one interface (Strategy)

Risk: a failed emergency text alerts the contact centre

Trade-off: more classes to learn and trace

NON-TECHNICAL TALK: FOR TENANTS

'You choose how we contact you: text, email or letter.'

Helps tenants like Joan and Amir, without phoning us

You hear when a repair is booked or moved

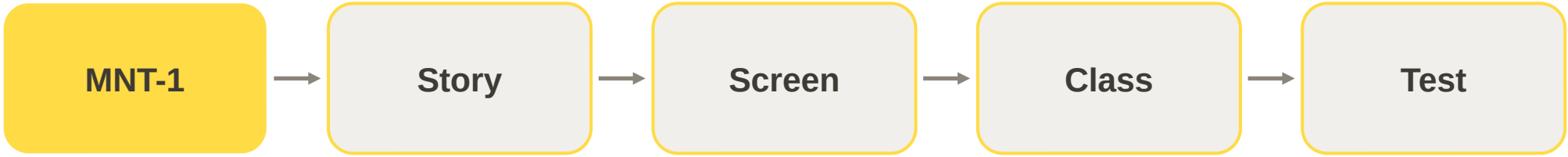
If an urgent text fails, an adviser is told at once

We need your up-to-date contact details

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One requirement, traced through the design

Worked example: one row of a traceability table. Your table needs three rows like it, for your own feature.



MNT-1	Story	Screen	Class	Test
A new channel needs no change to repair logic	Tenant chooses text, email or letter for updates	Contact preferences	NotificationChannel interface	Add a dummy channel; every repair test still passes

Build each row from the requirement rightwards. An empty cell is the first thing a reviewer will spot.

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Your review board

Role cards are in Supporting Material section 5. Play the role, not yourself.

Tenant panel member
External. Is it clear and usable for every tenant?

Contractor supervisor
External. What do my people need, and not need?

Presenter

Developer
Is the structure sound, and can we test it?

Data protection officer
What personal data is collected, shared, kept?

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Activity 1: Build your design pack

Individual

Extend the design of one feature into a pack a review board can follow.

1. Traceability page: name your feature, write R1 to R3 and send them to the tutor by chat.
2. Analysis page: one user story per requirement, each with one Given/When/Then criterion.
3. Wireframes page: one phone screen for the main path, with labels and an error message.
4. Model page: a class diagram of 4 to 6 classes, or a short sequence diagram.
5. Trace R1 to R3 to a story, screen, class and test; add two security checks.
6. Outline a two-minute technical talk and a two-minute non-technical talk.

TIME
45 minutes working time

GROUPING
Individual

YOU NEED
Session 6 starter file
Your project, or Case Brief Part 6
Review criteria (Supporting)

OUTPUT AND SUCCESS
A pack that traces R1 to R3 from story to test, and an outline of two talks.

Debrief: Which requirement was hardest to trace all the way to a test, and why?

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Break

20 minutes

Save your draw.io file to your device now. After the break: the review board, so keep your pack and outline open.

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Activity 2: Design review board

Groups of 3 or 4, breakout rooms

Present to a mixed board, question in role, and give feedback the presenter can use.

1. Agree the order, a timekeeper and role cards, including an external role (2 min).
2. Presenter: share your screen; give your technical talk, then your non-technical talk (4 min).
3. Board: ask questions in role; the presenter answers and notes what to change (3 min).
4. Board: each write two specific points on the feedback form; post them in the chat (2 min).
5. Rotate. Groups of 3: 5 minutes of questions. Copy your feedback before the room closes.

TIME
40 minutes working time

GROUPING
Groups of 3 or 4, breakout rooms

YOU NEED
Your pack and talk outline
Role cards (Supporting)
Feedback form (Activities)

OUTPUT AND SUCCESS
Everyone presents once, gets written feedback and gives two specific points per presenter.

Debrief: Which question from an external role changed how you see your design?

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From feedback to a development goal

Worked example on a different design. Your goal must come from your own feedback.

ANOTHER LEARNER'S LOG ENTRY

Feedback: 'Your sequence diagram shows no path for a failed payment.'

(1) Change: added the failed-payment path and a test for it

(2) Goal: draw the main error path for every feature I design at work

(3) Measure: my lead checks two designs and finds the error path in both

(4) Date: by 26 February 2027, at my skills coach review

1. A change you made, or plan to make, with a reason
2. Specific: what you will do, and where
3. Measurable: how you, or someone else, will know
4. Dated, with someone who will check

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Activity 3: Act on feedback

Individual

Turn the feedback you received into one design change and one development goal.

1. Read the feedback posted for you and choose the point that matters most (1 min).
2. Make the change in your design pack, or write the change you will make and why (4 min).
3. Write one development goal for your next module: specific, measurable, with a date (2 min).
4. Copy the feedback point, your change and your goal into the Session 6 log entry (1 min).

TIME
8 minutes working time

GROUPING
Individual

YOU NEED
Your board's feedback
Your design pack
Reflective Log.docx

OUTPUT AND SUCCESS
One change made or planned with a reason, and one dated, measurable goal in your log.

Debrief: Read out your goal. Could someone check on that date whether you met it?

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Check your understanding

CHECK

1. Which reply best suits a tenant panel member asking what your design means for her?

A It adds an event-driven status service in front of the existing repairs API.
B You can check each repair on your phone, and we text you when it changes.
C The tracking screens meet WCAG 2.2 AA and ask one question per page.
D Status is cached in its own table so that the slow repairs API copes.

2. Which comment on a peer's design is the most specific?

A The wireframes are really clear and well laid out. Nice work overall.
B Some of the security could be tighter, especially around photo uploads.
C Screen 2 shows the error only as a red border; add text saying what to fix.
D I would rethink the class diagram, as it seems a bit too complicated.

3. At the design stage, what is a design review mainly for?

A Catching gaps and agreeing the design while change is still cheap.
B Giving the team lead a score for each developer's design work.
C Proving the design is finished so it never needs to change again.
D Checking the code follows the team's naming and layout conventions.

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Skills covered today

KSB and official wording	In plain English	Where we covered it today
Part of K4: how to adapt appropriately to different audiences	Changing what you say for a tenant, a manager or a developer	Activity 2
Part of K6: how teams work effectively to produce software	How design reviews help a team build the right thing	Design reviews; live check
S15: communicate software solutions and ideas to technical and non-technical stakeholders	Presenting and defending your design to a mixed review board	Activity 2
B1: Works independently and takes responsibility	Building your own design pack and acting on feedback	Activities 1 and 3
Part of B4: Works collaboratively with a wide range of people in different roles; internally and externally	Reviewing designs with classmates who play tenants, contractors, managers and developers	Activity 2

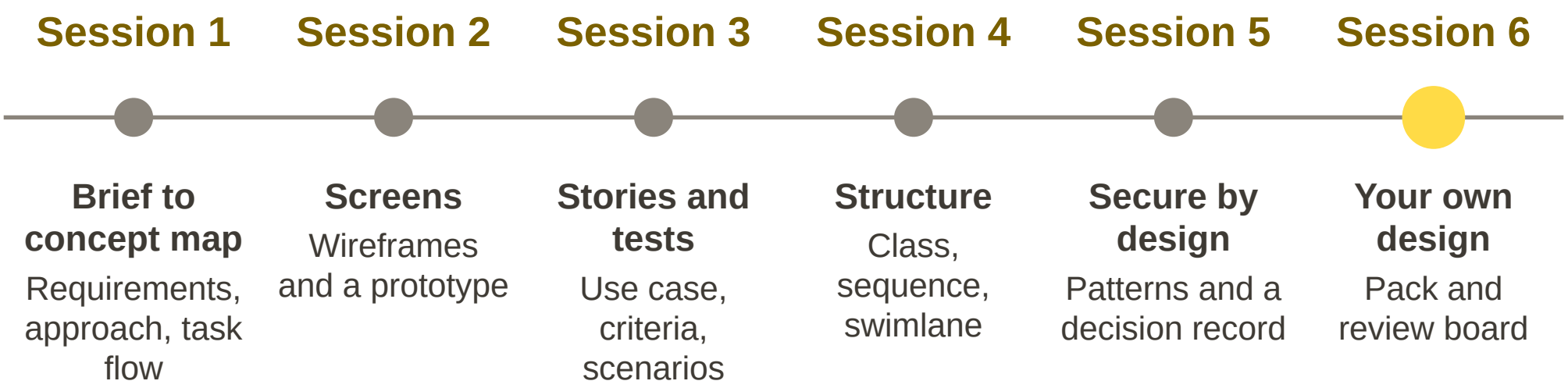
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Skills covered today (continued)

KSB and official wording	In plain English	Where we covered it today
B9: Committed to continued professional development.	Turning feedback into a development goal	Activity 3

From brief to reviewed design

Module 4 in six sessions: the same techniques, ending on your own project.



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What carries forward

Data & Databases

Class diagrams lead into entity relationship diagrams.

Classes that hold data become entities; attributes become columns; multiplicities become keys and relationships.

Data minimisation decides which columns exist at all.

Software Development

You implement designs like these in code, securely.

Server-side validation, role-scoped data and loud failures move from the diagram into code and tests.

Your traceability table becomes a starting test plan.

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Recap

1. A design pack explains one feature, traced from requirement to test.
2. Reviews catch gaps while change is cheap, if people feel safe to speak.
3. Specific feedback names the place, the criterion and a change.
4. One design, two talks: how it works, and what changes for people.
5. Feedback becomes a dated, measurable development goal.

Before your next module

Do the Session 6 Consolidate quiz: ten questions, about 10 minutes.

Finish your Reflective Log, including the End of module rows.

Keep your design pack and feedback; it can be portfolio evidence if your skills coach agrees. Read the Prepare document for your next module.

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Reflective Log

About three minutes, now

Complete the Session 6 entry in Reflective Log.docx: the feedback you received, the change you made and your development goal.

Fill in the End of module rows: the design technique you will use first at work, and one thing you will practise before your next module.

Then copy the key points into your OneFile timesheet reflection so your skills coach can see them.

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